will discuss the origin of this blurring, but before I do, let's take another step by bringing quantum mechanics into consideration.

QUANTUM TIME

Roger Penrose is among the most lucid of those scientists who have focused on space and time.[83] He reached the conclusion that the physics of relativity is not incompatible with our experience of the *flowing* of time but that it does not seem sufficient to account for it. He has suggested that what's missing might be what happens in a quantum interaction.[84] Alain Connes, the great French mathematician, has pointed out the deep role of quantum interaction at the root of time.

When an interaction renders the *position* of a molecule concrete, the state of the molecule is altered. The same applies for its *speed*. If what materializes *first* is the speed and *then* the position, the state of the molecule changes *in a different way* than if the order of the two events were reversed. The order matters. If I measure the position of an electron first and then its speed, its state changes differently than if I were to measure its velocity first and then its position.

This is called the "noncommutativity" of the quantum variables, because position and speed "do not commute," that is to say, they cannot exchange order with impunity. This noncommutativity is one of the characteristic phenomena of quantum mechanics. Noncommutativity determines an order and, consequently, a germ of temporality in the determination of two physical variables. To determine a physical variable is not an isolated act; it involves interaction. The effect of such interactions depends on their order, and this order is a primitive form of the temporal order.

Perhaps it is the very fact that the effect of these interactions depends on the order in which they take place that is at the root of the temporal order of the world. This is the fascinating idea suggested by Connes: the first germ of temporality in elementary quantum transitions lies in the fact that these interactions are naturally (partially) ordered.

Connes has provided a refined mathematical version of this idea: he has shown that a kind of temporal flow is implicitly defined by the noncommutativity of the physical variables. Due to this noncommutativity, the set of physical variables in a system defines a mathematical structure called "noncommutative von Neumann algebra," and Connes has shown that these structures have within themselves an implicitly defined flow.[85]